
GENERAL RELATIVITY TUTORIAL

Problem set 3, 27.09.2019.

■ PROBLEM 7 Surfaces and manifolds.

Consider the following surfaces imbedded in 3-dimensional Euclidean space: (i) a sphere, (ii) a torus, (iii) surface of a cone, shown in Figure 2.

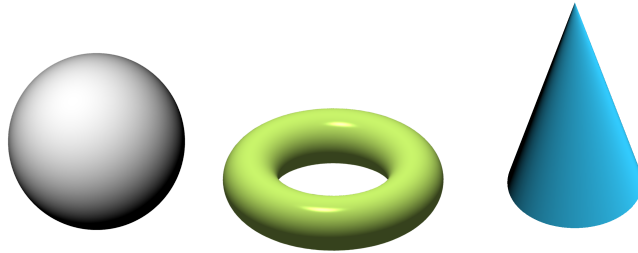


Figure 1: From left to right: a sphere, a torus, a cone.

- (a) Construct a set of charts (atlas) for each of these surfaces, i.e. construct coordinate systems on these surfaces by expressing them in terms of 3-dimensional Cartesian coordinates. What can you say about the tip of the cone (in the context of differentiable manifolds)?
- (b) Find what an infinitesimal line element is on these surfaces (in one particular chart).
Hint: Remember that for 3-dimensional Euclidean space the line element in Cartesian coordinates is just $dl^2 = dx^2 + dy^2 + dz^2$.

■ PROBLEM 8 Coordinate transformations.

- (a) How would the product of a covariant and a contravariant vector, $A^\mu B_\mu$, transform under general coordinate transformations?
- (b) What about the quantity $g_{\mu\nu} g^{\nu\rho} = \delta_\mu{}^\rho$?
- (c) Can you show that the derivative of a covariant vector, $\partial_\mu A_\nu$, does not transform as a tensor? Show that for linear coordinate transformations (*i.e.* Lorentz transformations) $\partial_\mu A_\nu$ is indeed a tensor.
- (d) Check that the transformation rules (4.10) and (4.13) in the lecture notes (http://www.staff.science.uu.nl/~hooft101/lectures/genrel_2013.pdf) form a group, i.e. the transformation $x \rightarrow u$ yields the same tensor as the sequence $x \rightarrow v \rightarrow u$. Make use of the fact that partial differentiation obeys

$$\frac{\partial x^\mu}{\partial u^\nu} = \frac{\partial x^\mu}{\partial v^\alpha} \frac{\partial v^\alpha}{\partial u^\nu} . \quad (8.1)$$

■ **PROBLEM 9** (Anti)-symmetrizing tensors.

For any 2-tensor $T_{\mu\nu}$ (in four dimensions) we define the tensors

$$T_{(\mu\nu)} = \frac{1}{2}(T_{\mu\nu} + T_{\nu\mu}) , \quad T_{[\mu\nu]} = \frac{1}{2}(T_{\mu\nu} - T_{\nu\mu}) . \quad (9.1)$$

These are, respectively, called symmetric and antisymmetric part of the tensor $T_{\mu\nu}$.

(a) Is it true that for any tensor $A_{\mu\nu} = A_{(\mu\nu)} + A_{[\mu\nu]}$? How many components do $A_{(\mu\nu)}$ and $A_{[\mu\nu]}$ have?

(b) If $B_{\mu\nu} = B_{\nu\mu}$ and $C_{\mu\nu} = -C_{\nu\mu}$, prove that (A is a generic tensor)

$$A_{\mu\nu}B^{\mu\nu} = A_{(\mu\nu)}B^{\mu\nu} , \quad A_{\mu\nu}C^{\mu\nu} = A_{[\mu\nu]}C^{\mu\nu} . \quad (9.2)$$

For higher rank tensors, this concept can be generalized to obtain totally symmetric and anti-symmetric tensors by definitions

$$T_{(\mu_1\mu_2\ldots\mu_k)} = \frac{1}{k!} \sum_{\sigma \in S^k} T_{\sigma(\mu_1)\sigma(\mu_2)\ldots\sigma(\mu_k)} , \quad (9.3)$$

$$T_{[\mu_1\mu_2\ldots\mu_k]} = \frac{1}{k!} \sum_{\sigma \in S^k} (-1)^{\pi(\sigma)} T_{\sigma(\mu_1)\sigma(\mu_2)\ldots\sigma(\mu_k)} , \quad (9.4)$$

respectively, where T is an arbitrary covariant tensor of rank k . In (9.3) and (9.4), S^k denotes the group of k objects, and π the so-called parity of the permutation, with $\pi(\sigma) = 0$ if σ is an even permutation, and $\pi(\sigma) = 1$ if it is an odd permutation.

(c) Consider the case of an arbitrary 3-tensor $T_{\mu\nu\rho}$. How many components does it have? Write explicitly the form of $T_{(\mu\nu\rho)}$ and $T_{[\mu\nu\rho]}$.

(d) For an arbitrary 3-tensor $T_{\mu\nu\rho}$ is it true that $T_{\mu\nu\rho} = T_{(\mu\nu\rho)} + T_{[\mu\nu\rho]}$? Prove your result.